PHY 521: Stars

Homework #2

Due: 2025-09-23 (by the end of the scheduled class period)

Be sure to show all work. Assignment is turned in via Brightspace assignment upload. You should make a single, legible PDF with your solutions. You must show your work and include units with all numbers, all throughout each problem.

While you are free to discuss problems with your classmates, your work must be your own. Any use of AI/LLM must be disclosed. All calculations (including any code you may write) and writing up of the problem solution must be your own work.

Unless otherwise noted, all problems carry equal weight.

1. *Degeneracy.* In class we showed that

$$P_e = A f(x) \tag{1}$$
$$E_e = A g(x) \tag{2}$$

with

$$A = \frac{\pi}{3}\left(\frac{m_e c}{h}\right)^3 m_e c^2 \tag{3}$$

and

$$f(x) = x(2x^2 - 3)(1 + x^2)^{1/2} + 3\sinh^{-1} x \tag{4}$$
$$g(x) = 8x^3\left[(1 + x^2)^{1/2} - 1\right] - f(x) \tag{5}$$

Here, x is the Fermi momentum, $x = p/(m_e c)$.

(a) Show that in the limit $x \ll 1$, that we get

$$P_e \propto (\rho/\mu_e)^{5/3} \tag{6}$$
$$E_e \propto (\rho/\mu_e)^{5/3} \tag{7}$$

by expanding $f(x)$ and $g(x)$ for $x \ll 1$, and find the proportionality constants. Evaluate these constants in CGS units, i.e., you should write it as

$$P_e = K\left(\frac{\rho/1\ \mathrm{g\ cm^{-3}}}{\mu_e}\right)^{5/3} \tag{8}$$

where K has units of $\mathrm{dyn/cm^2}$.

(b) Do the same for the relativistic limit, $x \gg 1$, where

$$P_e \propto (\rho/\mu_e)^{4/3} \tag{9}$$
$$E_e \propto (\rho/\mu_e)^{4/3} \tag{10}$$

2. *Non-zero temperature degeneracy.* In class we looked at the limit of zero temperature and computed the pressure and energy density of an electron gas for arbitrary degree of relativity. This approximation was essentially taking the degeneracy parameter, $\eta = \mathcal{E}_F/kT \to \infty$.

See our class notebook for a review of the degeneracy parameter: `https://zingale.github.io/stars/notebooks/eos/degeneracy-parameter.html`.

The advantage of this approximation was that we could do the integrals analytically.

Now we'll consider another approximation that can be done analytically: non-zero temperature and non-relativistic. This means that η is large but not infinite.

(a) Show that the number density of electrons, n_e, and pressure, P_e, can be written as

$$n_e = \frac{4\pi}{h^3}(2m_e kT)^{3/2} F_{1/2}(\eta) \tag{11}$$

$$P_e = \frac{8\pi}{3h^3}(2m_e kT)^{3/2}(kT) F_{3/2}(\eta) \tag{12}$$

where the *Fermi-Dirac* integral is defined as:

$$F_n(\eta) = \int_0^\infty \frac{\xi^n}{e^{\xi-\eta}+1} d\xi \tag{13}$$

Hint: you will want to change your integration variable to be in terms of energy, $\mathcal{E}$, instead of momentum and use the non-relativistic energy and momentum.

(b) For large η, an expansion of the Fermi-Dirac integral using the *Sommerfield expansion* is

$$F_n(\eta) = \frac{\eta^{n+1}}{n+1}\left[1 + \frac{\pi^2}{6}(n+1)n\eta^{-2} + \mathcal{O}(\eta^{-4})\right] \tag{14}$$

(see Chandrasekhar 1939 or Kippenhahn & Wiegert).

Use this expansion to leading order in η in our number density expression to find an expression for η in terms of number density and temperature, $\eta = \eta(n_e, T)$.

(c) Now approximate the pressure integral by keeping the first two terms of the $F_{3/2}(\eta)$ expansion, and show that we get an electron pressure of the form:

$$P_e = \frac{h^2}{20m_e}\left(\frac{3}{\pi}\right)^{2/3} n_e^{5/3}\left[1 + \alpha\frac{T^2}{n_e^{4/3}}\right] \tag{15}$$

Find the expression for α.

Notice that the leading order term is exactly what we found for non-relativistic degeneracy pressure at zero temperature in class and the next term is the correction with temperature. If that term becomes big, then the degeneracy is "lifted".

(d) *Adiabatic Index*. The quantity:

$$\Gamma_1 = \left.\frac{d\log P}{d\log\rho}\right|_s = \left.\frac{\rho}{P}\frac{dP}{d\rho}\right|_s \tag{16}$$

plays an important role in stellar structure (e.g. this is the "gamma" that appears in the sound speed). Here s is the specific entropy, which is held constant.

Consider a mix of radiation and gas, with the pressure given by:

$$P = P_\gamma + P_g = \frac{1}{3}aT^4 + \frac{\rho kT}{\mu m_u} \tag{17}$$

and define β as:

$$\beta = \frac{P_g}{P} \tag{18}$$

Find an expression for Γ_1 for this equation of state in terms of β, i.e. $\Gamma_1 = f(\beta)$.

Hint: One way to proceed is to start with the pressure written as $P = P(\rho, T(\rho, s))$, and then compute $dP/d\rho|_s$. You will need to find $dT/d\rho|s$, which you can find via the first law of thermodynamics:

$$Tds = 0 = de + Pd(1/\rho) \tag{19}$$

and writing $de = \partial e/\partial\rho|_T\, d\rho + \partial e/\partial T|_\rho\, dT$. Your result should give the expected values as $\beta \to 0$ and $\beta \to 1$.